

CodeAlpha Cyber Security Internship

Task 3: Secure Coding Review

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Introduction

This project reviews a Python-based login application to identify security vulnerabilities and implement secure coding practices.

Application Overview

The application uses Python and SQLite to authenticate users using a username and password.

Vulnerabilities Identified

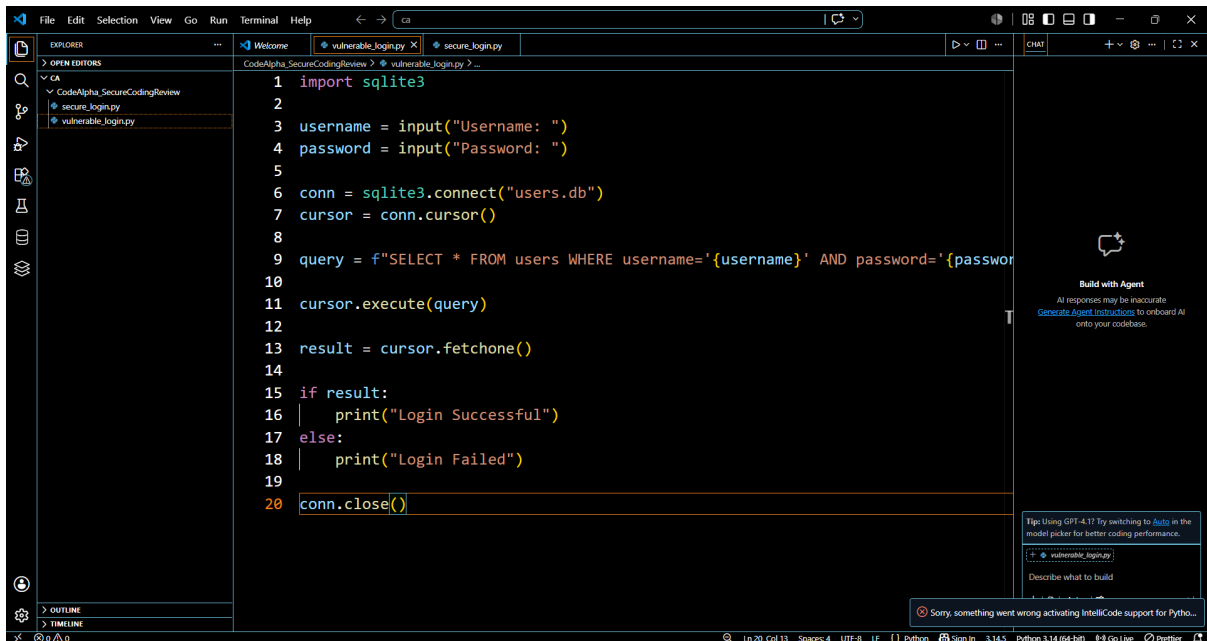
1. SQL Injection (High)
2. Plain Text Password Storage (High)
3. Missing Input Validation (Medium)
4. Missing Error Handling (Medium)
5. Missing Logging (Low)

Remediation

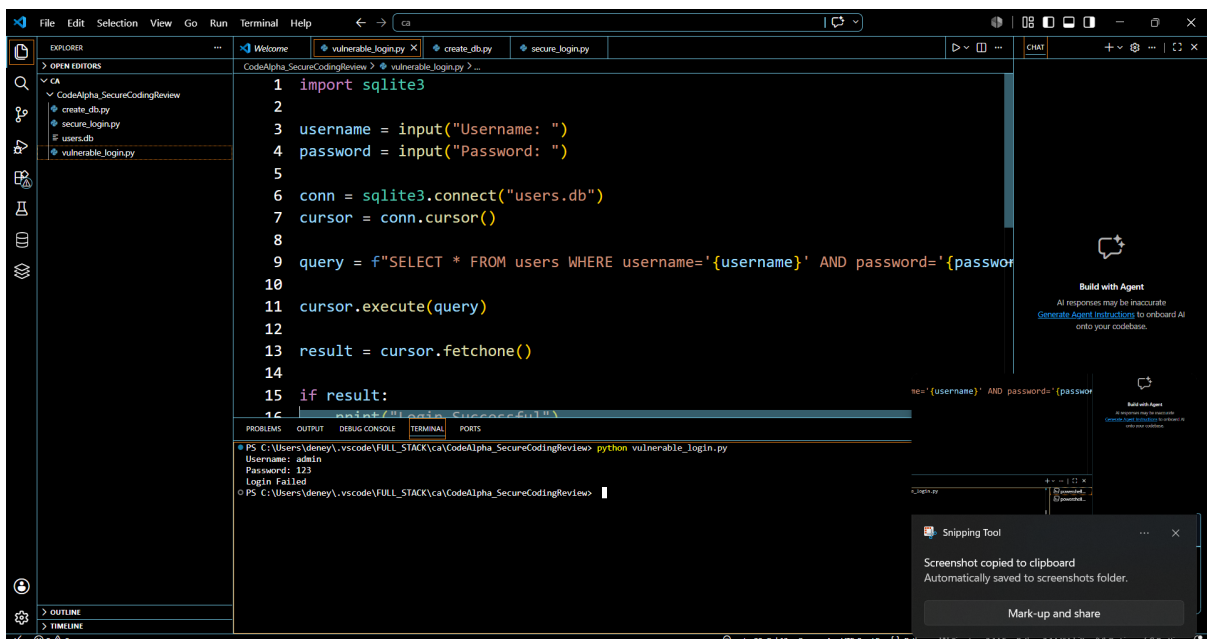
- Replaced string concatenation with parameterized queries.
- Added exception handling using try-except blocks.
- Recommended password hashing using bcrypt.
- Recommended input validation.
- Recommended authentication logging.

Screenshots

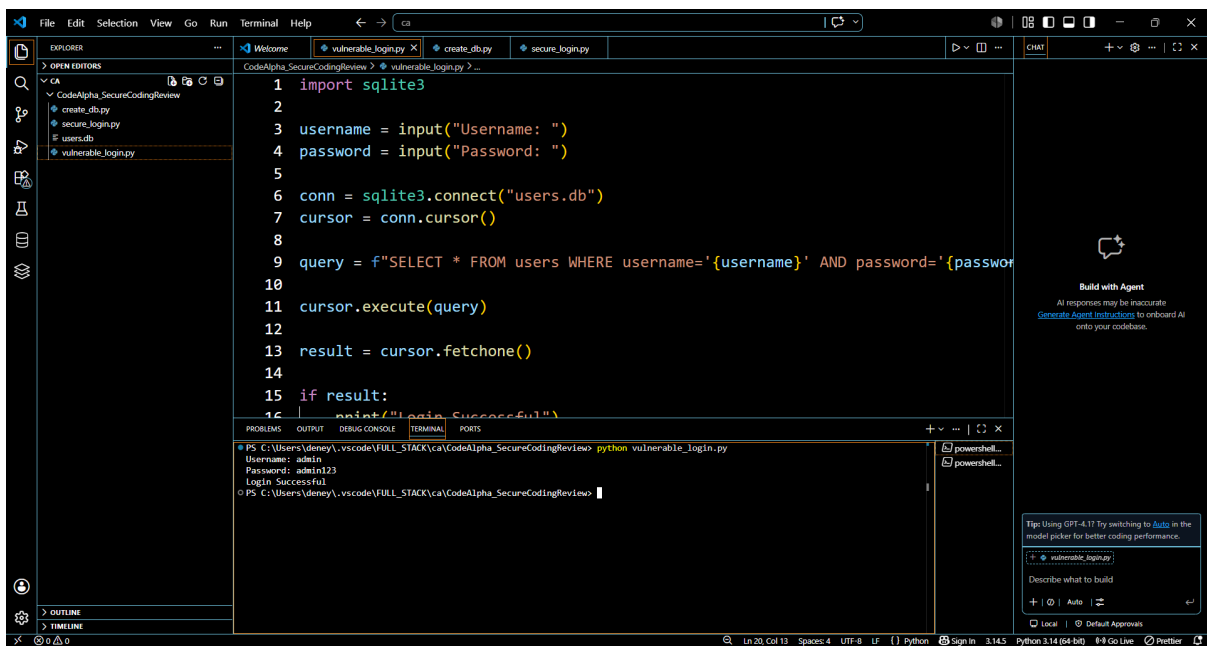
Screenshot 1



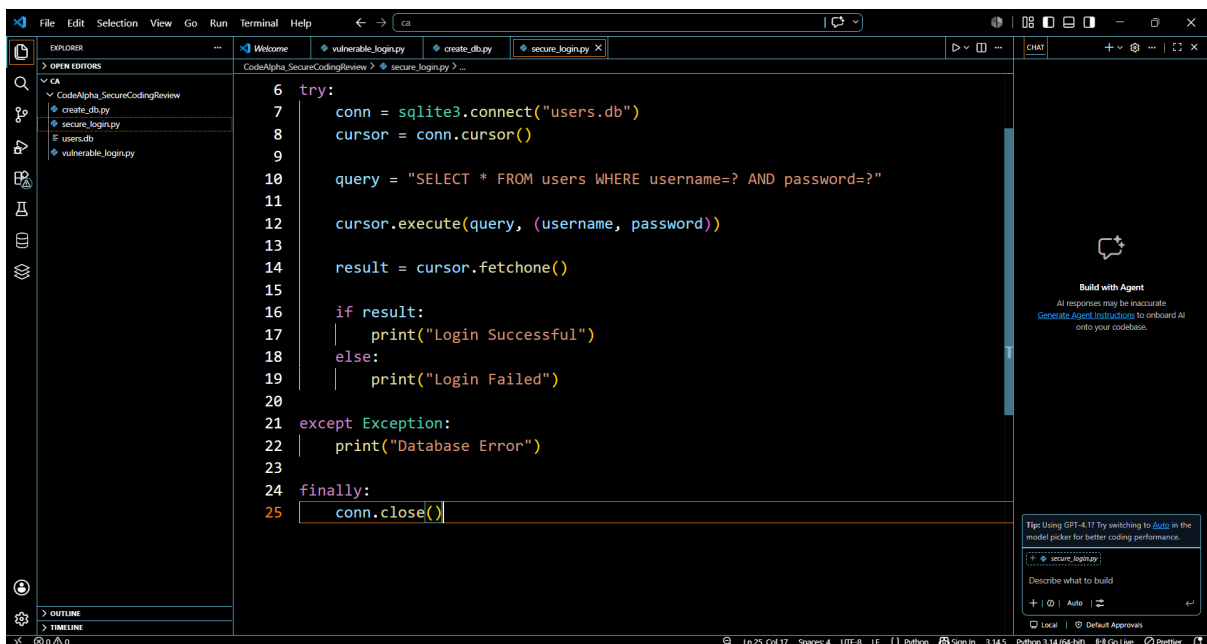
Screenshot 2



Screenshot 3



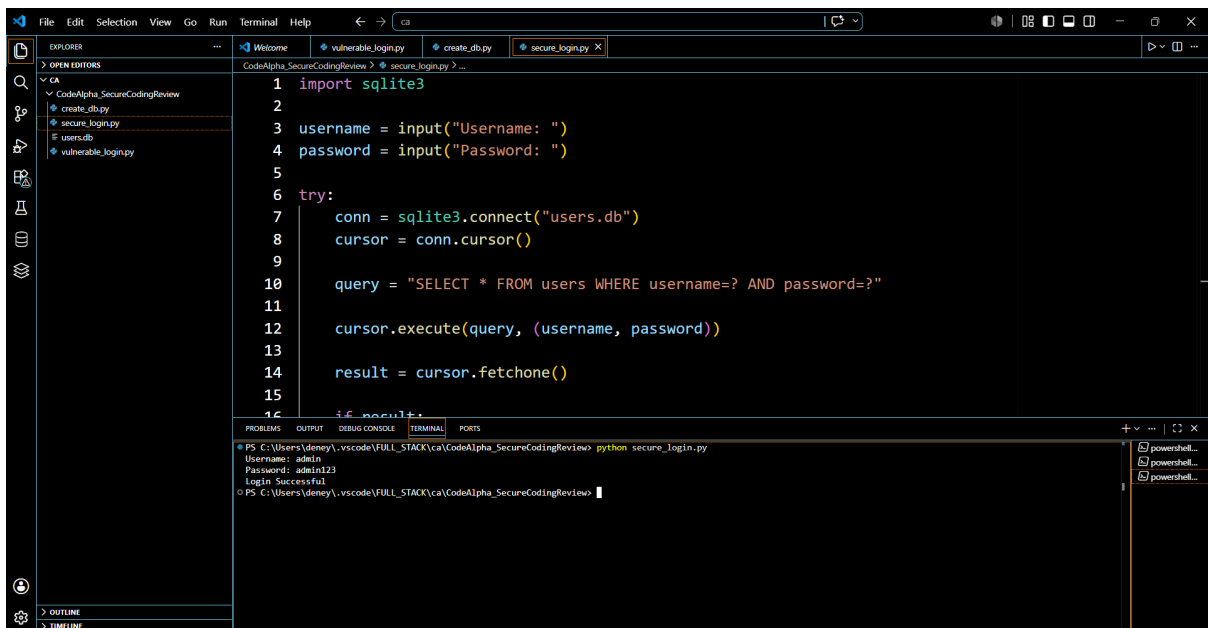
Screenshot 4



Screenshot 5



Screenshot 6



Conclusion

The secure coding review identified multiple vulnerabilities in the login application. The most critical issue was SQL Injection. A secure version was developed using parameterized queries and exception handling, significantly improving the application's security.